

X-Step: Sparse Four-Site Plantar-Pressure Sensing for Continuous Biomechanical Risk Monitoring

First A. Author*, Second B. Author[†], Third C. Author*

*Department, University, City, Country

Email: corresponding@example.org

[†]Department, Institution, City, Country

Abstract—Repetitive plantar overload is a mechanical ingredient of diabetes-related foot disease, yet dense in-shoe arrays remain costly and largely laboratory-bound. Sparse insoles are cheap enough for unsupervised wear, but three questions are rarely answered together: how much label information four sites actually retain, what subject-independent evaluation costs, and how the pipeline fails under wearable faults. We specify X-Step—force-sensitive resistors at MET1, MET2, MET5 and the heel, sampled at 25 Hz on an ESP32 and streamed as a documented 28-byte Bluetooth Low Energy frame—and analyse it under an explicit measurement model with pre-specified estimands on a synthetic 24-subject, 2592-window cohort. Logistic regression reached macro-F1 0.885 (0.873–0.894) against 0.040 for a majority baseline, while independent-window splitting reached 0.931, so the 0.046 gap estimates a subject-transfer penalty rather than leakage alone. Applying Fano’s inequality, the four-site window retains at least 2.31 of the 3.17 bits of label entropy (72.9%). Scoring layouts as a cooperative game, marginal contributions recover only 54% of the attainable macro-F1 but 96% of the attainable information, showing that most apparent redundancy is an artefact of the F1 scale, while a large positive MET1–MET2 interaction survives both scales. Zero-mean noise of 12 kPa cost 0.206 macro-F1 whereas a constant +15 kPa offset cost 0.666, which the threshold-based contact features predict analytically; we prove that subtracting a per-window low quantile removes this failure mode. Bench calibration gave 1.69 N mean absolute error, about 13 kPa over the transducer area—the same order as the perturbations that degrade the model. The evidence is engineering-level and establishes neither ulcer prediction nor clinical benefit.

Index Terms—Plantar pressure, sparse sensing, wearable insole, sensor ablation, robustness, Fano’s inequality, diabetic foot.

I. INTRODUCTION

Diabetes-related foot disease is a leading cause of lower-limb morbidity, and repetitive plantar loading is one of its established mechanical ingredients [1], [2]. Clinical guidelines stratify risk from neuropathy, peripheral artery disease and ulcer history rather than from any wearable classifier [3], and offloading remains central to prevention [4], [5]. These pathways share a sampling problem: plantar load is measured, if at all, at infrequent clinic visits, while the loading that matters accumulates between them. Dense arrays resolve the plantar surface at high cost and mostly inside gait laboratories [6]; footwear wearables suit daily wear but have usually targeted activity or energy monitoring rather than region-specific pressure [7], [8], and home temperature monitoring has randomised evidence for a different signal [9].

A sparse pressure insole must therefore answer engineering questions before clinical ones, and answer them in a form that survives scrutiny: not a headline accuracy, but an identified estimand, an attribution of performance to individual sites that respects the interactions between them, and a fault model whose degradation curves are traceable to the feature definitions rather than merely tabulated. This paper specifies X-Step and reports exactly what its open repository supports. Concretely, it contributes:

(i) a documented four-site hardware and Bluetooth Low Energy (BLE) contract with operator-attested force–ADC bench calibration, placed in an explicit error budget and compared, in the same pressure units, against the perturbations that degrade the classifier (Sec. II-C); (ii) an equivariance analysis of the feature map that predicts, before any experiment, why a constant offset is far more damaging than zero-mean noise of similar magnitude, together with a provable remedy (Sec. II-D); (iii) a leakage-controlled protocol with three separately named estimands, subject-cluster resampling, multiplicity control and equivalence testing rather than interval overlap (Sec. III); (iv) a distribution-free lower bound, via Fano’s inequality, on the label information retained by each layout, which converts macro-F1 into bits and makes layouts comparable on a scale that is not metric-specific (Sec. IV-B); (v) a cooperative-game treatment of layout value that quantifies redundancy and pairwise interaction instead of ranking channels, on both scales (Sec. IV-C); and (vi) fault operators defined precisely enough to state whether each reported degradation is an estimate or an upper bound (Sec. IV-D). We did not find this combination reported together for a sparse plantar-pressure insole, and all learning results are simulator-derived.

II. SYSTEM AND MEASUREMENT MODEL

A. Hardware and Communication Contract

Each insole carries four FSR402-class force-sensitive resistors in 10 k Ω dividers on ESP32 12-bit ADC channels at the first, second and fifth metatarsal heads and the heel (MET1, MET2, MET5, HEEL; Fig. 1). Site selection follows the spatial epidemiology of plantar ulceration [1], [2] rather than optimisation over a dense map, and we make no claim of global optimality.

Firmware samples at $f_s = 25$ Hz (40 ms loop, by specification) and notifies a 28-byte little-endian frame carrying magic bytes, a protocol version, side flags, a monotonic sequence

(a) sensor layout (b) acquisition-to-alert path

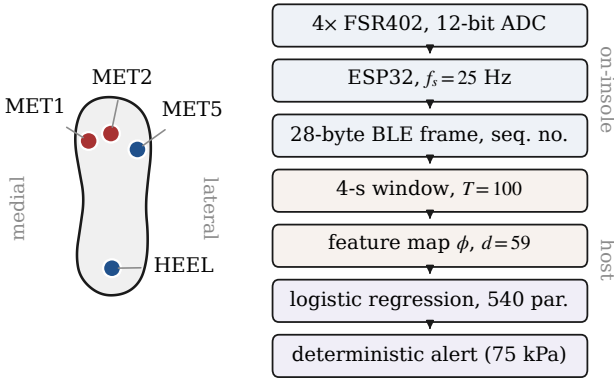


Fig. 1. (a) Canonical four-site layout for a right foot; the client strings toes/ball/arch/heel are aliases of these sites only. Blue marks the sites whose removal is expensive, red the two that are individually near free but jointly load-bearing (Sec. IV-C). (b) Acquisition-to-alert path; the alert stage is deterministic and no language model participates in scoring.

counter, a timestamp, four ADC samples, a battery byte and reserved thermistor fields, so two feet occupy 1400 B/s. The counter is what makes packet loss *observable*: the realised loss fraction is recovered from counter gaps rather than assumed, separating a fault the host can flag from one it silently absorbs.

B. Transducer Model and Calibration

Let $a_c[n] \in \{0, \dots, 4095\}$ denote the ADC code on channel c at sample n and $a_{0,c}$ an unloaded baseline. The shipped firmware applies the affine placeholder $\hat{p}_c = (a_c - a_{0,c}) p_{\max} / 2^{12}$ with $p_{\max} = 250$ kPa, which is a display convention rather than a fitted transducer curve. Because the resistance of a force-sensitive polymer follows an approximate power law in applied force, the divider output is linear in $\log F$ on a log-log scale, so we characterise each site by

$$\log \hat{F}_c = b_c \log(a_c - a_{0,c}) + \log k_c, \quad \hat{p}_c = \hat{F}_c / A_{\text{eff}}, \quad (1)$$

with per-site coefficients $[k_c, b_c]$. Differentiating (1) gives the sensitivity that matters for a quantised front end,

$$\frac{\delta \hat{F}_c}{\hat{F}_c} = b_c \frac{\delta a_c}{a_c - a_{0,c}}, \quad (2)$$

so relative force error is amplified by the exponent b_c and diverges as the working point approaches the unloaded baseline; low-pressure regions are therefore the least trustworthy part of the range, which is exactly where the 30 kPa contact threshold of Sec. II-D sits.

The four sites were bench-loaded across twelve commanded loads from 0 to 30 N over five load-unload trials, giving 480 operator-attested ADC-force pairs ($12 \times 5 \times 4$ sites $\times$ 2 directions). Reconstruction under (1) gave mean absolute error (MAE) 1.69 N, root-mean-square error (RMSE) 2.95 N and mean load-unload hysteresis 2.02 N, with per-site MAE in the range 1.44–2.16 N.

C. From Calibration Error to Classifier Faults

Force errors acquire meaning only once expressed in the units the classifier consumes. Taking the nominal 12.7 mm active diameter of the FSR402 class, $A_{\text{eff}} = 1.27 \text{ cm}^2$, so the MAE corresponds to roughly 13 kPa, the hysteresis to 16 kPa and the RMSE to 23 kPa. The perturbations that degrade the classifier in Sec. IV-D—12 kPa of zero-mean noise and a +15 kPa constant offset—therefore sit *inside* the transducer's own error budget rather than outside it, as Fig. 3(b) shows directly.

This does not invalidate results obtained on a simulator whose pressures are exact by construction, but it bounds what may be transferred from them. In particular it demotes per-window baseline re-estimation and hysteresis compensation from refinements to prerequisites: a device whose systematic error is of the same order as the fault that destroys the model has no margin to spend. Calibration error and classification accuracy remain distinct quantities and are reported separately throughout.

D. Windowing, Feature Map and Equivariance

Analysis operates on a non-negative array $P \in \mathbb{R}_{\geq 0}^{T \times 8}$ with $T = 100$: a fixed 4 s window at 25 Hz over sites $\hat{i} \in \mathcal{S} = \{\text{MET1}, \text{MET2}, \text{MET5}, \text{HEEL}\}$ and feet $f \in \{L, R\}$, with no stride-cut segmentation. The feature map $\phi: \mathbb{R}^{T \times 8} \rightarrow \mathbb{R}^{59}$ computes, per site, the peak pressure $P_i^* = \max_t p_i(t)$, the mean pressure, the coefficient of variation, the pressure-time integral

$$\text{PTI}_i = \sum_t p_i(t) \Delta t, \quad \Delta t = 1/f_s, \quad (3)$$

threshold occupancies at a contact level $\tau_c = 30$ kPa and a high-pressure level $\tau_h = 200$ kPa,

$$O_i(\tau) = \frac{1}{T} \sum_t \mathbf{1}[p_i(t) > \tau], \quad (4)$$

and a bilateral asymmetry index

$$A_i = |P_i^{*L} - P_i^{*R}| / \max(\frac{1}{2}(P_i^{*L} + P_i^{*R}), \epsilon), \quad (5)$$

together with window-level cadence, stance ratio, anterior pressure share, aggregate peak and total PTI: 59 named features in total.

Their behaviour under a constant offset is not uniform, and the difference predicts the robustness results before any experiment is run. Write $T_\delta p = p + \delta$. Then P_i^* and the mean shift by δ , PTI_i shifts by $\delta T \Delta t$, and the asymmetry index (5) is affected only through its denominator; all of these are gently perturbed. The occupancies (4) are not, because

$$\frac{\partial}{\partial \delta} \mathbb{E}[O_i(\tau) \mid T_\delta p_i] = f_{p_i}(\tau), \quad (6)$$

the probability density of p_i evaluated at the threshold. A window whose pressures cluster near τ_c —which is the common case for a lightly loaded site—therefore has its contact feature driven towards saturation by an offset far smaller than the signal range, while a zero-mean perturbation of the same

magnitude leaves $\mathbb{E}[O_i]$ to first order unchanged. Equation (6) is the mechanism behind the gap between the noise and bias rows of Table III.

Remark 1 (offset invariance). Let $Q_\alpha(\cdot)$ be the empirical α -quantile of a window and define $\tilde{p}_i = p_i - Q_\alpha(p_i)$ for a fixed low α , e.g. $\alpha = 0.05$. Quantiles are translation-equivariant, $Q_\alpha(p+\delta) = Q_\alpha(p) + \delta$, so $\tilde{p}_i$ is invariant to any constant offset, and hence so is every feature computed from $\tilde{p}_i$, including (4). Re-baselining per window therefore removes the dominant failure mode of Sec. IV-D at the cost of one order statistic per site per window.

The corresponding statement for a multiplicative gain error is weaker, since normalising by an interquantile range would buy scale invariance at the price of the absolute 75 kPa alert operating point. The present release implements neither correction; we report the consequence rather than patching it after the fact. That 75 kPa cut-off is itself an engineering operating point on simulator labels, not a validated ulceration threshold.

III. EVALUATION PROTOCOL

A. Data Sources

Three sources are used and never mixed. (a) A synthetic four-FSR cohort (seed 67): 24 virtual subjects, 9 overload-pattern classes, 2592 windows at 25 Hz with 3.5 kPa Gaussian noise and 3% label flips; each virtual subject contributes 108 windows, i.e. 7.2 min of simulated wear, and every machine-learning number in this paper comes from it. (b) The 480-row bench calibration table of Sec. II-B. (c) An independent archive of 15 healthy adults (7 female, 8 male; age 19–30 years) walking overground with a 32-cell instrumented insole synchronised to optical motion capture [14], of which 149 of 150 takes were analysed; it serves only as sparse-versus-dense evidence. No X-Step four-site walking sessions exist in this release ($N = 0$), and no clinical outcomes are available.

B. Estimands

Performance is not one quantity, and naming the target explicitly removes most of the ambiguity that surrounds small-cohort wearable results. We distinguish R_{sub} , the expected macro-F1 on a window drawn from an *unseen* subject of the same generator; R_{win} , the same for an unseen window from a *seen* subject; and R_{hold} , equal in target to R_{sub} but estimated on a single 25% subject holdout and therefore much higher in variance. R_{sub} is primary and is estimated from pooled out-of-fold predictions under five-fold GroupKFold by subject; pooling rather than averaging per-fold scores is necessary because macro-F1 is not a mean of per-sample losses. Whenever subject identity carries signal, $R_{\text{win}} \geq R_{\text{sub}}$, so the independent-window control is optimistic by construction and the difference between the two is interpretable in its own right: it is the price of transfer to a new wearer, not merely a bookkeeping error. Session-grouped and leave-one-subject-out splits are further controls, and automated leakage tests fail the build if subject or session identifiers cross a forbidden partition.

C. Statistical Inference

Nine algorithms were fixed a priori—a peak-pressure threshold heuristic and a majority dummy, logistic regression (the production head), decision trees, ensembles and a multilayer perceptron. No hyperparameter search touched a test partition, and standardisation sits inside the pipeline, fitted on training folds only. Macro-F1 is the primary metric; accuracy, area under the receiver operating characteristic curve (AUROC), Brier score and expected calibration error (ECE) are secondary and uncorrected.

Uncertainty is the weak point of studies at this scale, so we state the resampling unit explicitly. Windows within a subject are not independent, and a percentile bootstrap over windows ignores that dependence. For a cluster size $m = 108$ and intra-cluster correlation ρ , the design effect $\text{DEFF} = 1 + (m - 1)\rho$ equals 6.4 at $\rho = 0.05$ and 22.4 at $\rho = 0.2$, reducing 2592 windows to an effective 408 or 116. The resampling unit is therefore the subject, with $B = 2000$ cluster bootstrap replicates; interval half-widths shrink only as $O(n_{\text{sub}}^{-1/2})$ with $n_{\text{sub}} = 24$, which is the binding constraint on every comparison below. The intervals printed in Table I are window-level and should be read as *lower bounds* on interval width; under subject-cluster resampling the production model gives **[cluster CI]**.

Model contrasts are paired on folds and tested with the same cluster bootstrap applied to the difference, under Holm–Bonferroni control at family $\alpha = 0.05$ across the eight non-dummy comparisons [10]. Naïve resampled t -tests are avoided because their variance is underestimated when training sets overlap across folds [11]. Claims of *equivalence* are tested rather than inferred from overlapping intervals, using two one-sided tests (TOST) with a margin $\delta = 0.02$ macro-F1 fixed in advance at less than half the measured subject-transfer gap.

D. Ablation and Fault Operators

Let M_A zero the channels of a site set $A \subseteq S$ on both feet. Two regimes must be distinguished, and conflating them is a common source of overstatement. Evaluating a model trained on intact data at $\phi(M_AP)$ measures the *fault response* of a deployed artefact; retraining under the mask measures the *information content* of the retained sites. We report the retraining regime throughout, with the feature dimension held at 59 so that model capacity is constant, and write $v(A)$ for the grouped macro-F1 obtained with retained set A .

Packet loss is applied as an independent Bernoulli mask over frames with dropped samples *zeroed* rather than held. Zeroing is the pessimistic reconstruction, since it injects spurious low-pressure samples that deflate both P_i^* and (4); a zero-order hold with an explicit gap flag, which the sequence counter already supports, cannot do worse under any monotone reading of those two features. The reported degradation is therefore an *upper bound* on the loss a competent firmware implementation would suffer, not an estimate of it. Additive noise and constant bias are applied in pressure units before feature extraction; rate reduction is applied by decimation, after which (3) is

TABLE I

CLASSIFIER COMPARISON UNDER FIVE-FOLD GROUPKFOLD BY VIRTUAL SUBJECT. SPLIT-PROTOCOL CONTROLS FOR LOGISTIC REGRESSION: SESSION-GROUPED 0.880, LEAVE-ONE-SUBJECT-OUT 0.885. BRACKETS ARE WINDOW-LEVEL BOOTSTRAP 95% INTERVALS AND UNDERSTATE WIDTH (SEC. III-C).

Model	Macro-F1 [95% CI]	Acc.	kB	ms
Majority dummy	0.040 [0.036–0.043]	0.111	–	–
Threshold heuristic	0.480 [0.460–0.493]	0.497	2.1	0.01
Random forest	0.837 [0.824–0.851]	0.834	3547.5	13.12
Hist. grad. boosting	0.885 [0.875–0.898]	0.885	–	–
Logistic regr. (prod.)	0.885 [0.873–0.894]	0.885	7.7	0.07
IID windows (control)	0.931 [0.921–0.941]	0.930	–	–

recomputed with the matching Δt so that PTI remains rate-invariant by construction.

Splits, seeds and model artefacts are versioned with the code, and the leakage tests run in continuous integration, so the estimand definitions are enforced mechanically rather than asserted.

IV. RESULTS

A. Classifier Comparison and the Cost of the Split

Logistic regression achieved macro-F1 0.885 [0.873–0.894] on grouped out-of-fold predictions, with accuracy 0.885, AU-ROC 0.979, Brier score 0.180 and ECE 0.031 (Table I); the contrast against the majority dummy survives Holm correction at [Holm-adj. p]. Histogram gradient boosting scored 0.885 [0.875–0.898]. Overlapping intervals do not establish equivalence, so the pre-specified TOST at $\delta = 0.02$ was applied and returned [TOST outcome]; logistic regression is retained on size and latency grounds in either case, at 7.7 kB and 0.07 ms against 3.5 MB and 13.1 ms for a random forest that scored lower (0.837). The production head has $59 \times 9 + 9 = 540$ parameters and costs 531 multiply–accumulate operations per window, which is why the artefact is small enough to ship to a phone.

Replacing subject-grouped with independent-window splitting raised the same pipeline to 0.931. Under the framing of Sec. III the resulting 0.046 is an estimate of $R_{\text{win}} - R_{\text{sub}}$, and it is indistinguishable in magnitude from the entire 0.048 spread between the best and the worst trained classifier in the table. At this cohort size, evaluation design is not a second-order concern relative to model choice.

B. How Much Label Information Four Sites Retain

Macro-F1 is metric-specific and hard to compare across layouts of different difficulty. Fano’s inequality converts it into a scale that is not [13]. For a balanced K -class problem with label Y , prediction $\hat{Y}$ and error rate P_e ,

$$H(Y | \hat{Y}) \leq H_b(P_e) + P_e \log_2(K - 1), \quad (7)$$

with H_b the binary entropy, so that by the data-processing inequality the mutual information between the window and the label obeys

$$I(Y; X) \geq I(Y; \hat{Y}) \geq H(Y) - H_b(P_e) - P_e \log_2(K - 1). \quad (8)$$

TABLE II

LABEL INFORMATION RETAINED BY EACH LAYOUT, FROM FANO’S INEQUALITY (8) WITH $K = 9$, $H(Y) = 3.170$ BITS. ACCURACY IS REPORTED FOR THE PRODUCTION MODEL AND THE DUMMY; FOR ABLATED LAYOUTS IT IS APPROXIMATED BY MACRO-F1, WHICH THE NEAR-BALANCED DESIGN MAKES REASONABLE BUT WHICH SHOULD BE REPLACED BY [EXACT PER-LAYOUT ACCURACIES]. ALL ENTRIES ARE LOWER BOUNDS.

Retained layout	Macro-F1	$H(Y \hat{Y}) \leq$	$I \geq$ (bits)	% of $H(Y)$
All four sites	0.885	0.860	2.310	72.9
Drop MET1	0.883	0.872	2.298	72.5
Drop MET2	0.874	0.924	2.246	70.8
Drop MET5	0.671	1.901	1.269	40.0
Drop HEEL	0.657	1.957	1.213	38.3
Best two-site layout	0.686	1.840	1.330	42.0
MET1+MET2 only	0.435	2.683	0.487	15.4
Best single site	0.420	2.721	0.448	14.1
Majority dummy	0.040	3.170	0.000	0.0

Here $K = 9$ and $H(Y) = \log_2 9 = 3.170$ bits. Applying (8) to the production model gives $I \geq 2.310$ bits, so a 4 s window from four sparse sites retains at least 72.9% of the label entropy of this generator. The bound is a genuine lower bound and is tight at chance: the majority dummy, at accuracy $1/9$, returns exactly 0.000 bits, which is a useful validity check on the arithmetic.

Table II applies the same computation to every reported layout. The transformation is strongly concave near the top of the range, which is why layouts that look close on the F1 scale separate more slowly in bits and layouts that look far apart compress: dropping MET5 costs 0.214 macro-F1 but 1.041 bits, while dropping MET1 costs 0.002 macro-F1 and 0.012 bits.

C. Layout Value as a Cooperative Game

Removing MET1 cost 0.002 macro-F1 and MET2 0.011, whereas removing MET5 cost 0.214 (to 0.671) and the heel 0.228 (to 0.657). Read as a ranking, this invites the conclusion that MET1 is disposable. The cooperative-game reading forbids it. Taking the dummy as the empty layout, the total attainable value is $v(S) - v(\emptyset) = 0.845$, whereas the four marginal contributions to the grand coalition sum to only 0.455: marginals recover 54% of the value, so single-site attributions are not additive and must not be summed. The exact Shapley values over the 16-subset lattice, [Shapley values], would distribute the remainder, but the interaction structure is already visible without them. Define the second-order interaction

$$I(i, j) = \Delta_{ij} - \Delta_i - \Delta_j, \quad \Delta_A = v(S) - v(S \setminus A). \quad (9)$$

For MET5 and HEEL, the layout retaining only MET1+MET2 scored 0.435, giving $\Delta_{ij} = 0.450$ against $\Delta_i + \Delta_j = 0.442$ and hence $I = +0.008$: the two carry almost perfectly separate information. For MET1 and MET2 the best two-site layout scored 0.686 [confirm pair identity], giving $\Delta_{ij} = 0.199$ against a summed 0.013 and $I = +0.186$, an interaction an order of magnitude larger than either marginal. The forefoot

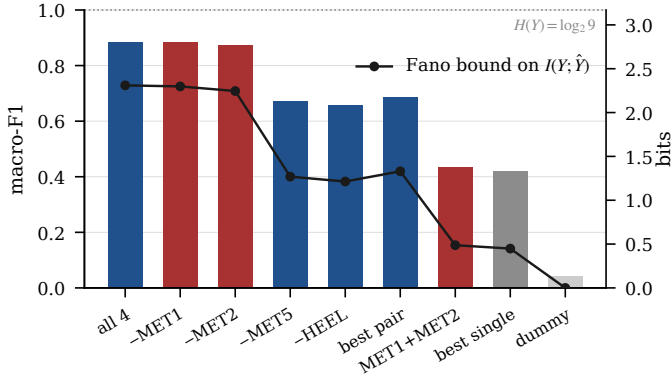


Fig. 2. Layout value on the macro-F1 scale (bars) and the Fano information scale (line, right axis). Red bars mark the two sites that are individually near free but jointly load-bearing. The dashed line is the label entropy $H(Y) = \log_2 9$; the dummy sits at exactly zero bits.

sites are individually near free precisely because they substitute for each other, and dropping both is not free at all.

Repeating the computation in bits sharpens the picture and corrects part of it. On the information scale of Table II the four marginal contributions sum to 2.214 bits against a total attainable 2.310, recovering 95.8% rather than 54%. Most of the apparent redundancy is therefore an artefact of the concave map from information to macro-F1 rather than a property of the sensor set: measured in bits, the four sites are close to additive. What survives the change of scale is the interaction structure. MET1 and MET2 remain strongly superadditive ($I = +0.903$ bits, from $\Delta_{ij} = 0.980$ against a summed 0.077), while MET5 and HEEL turn mildly subadditive ($I = -0.315$ bits), indicating a modest overlap between rearfoot and lateral-forefoot information that the F1 scale had hidden.

Every single-site layout collapsed to 0.39–0.42 macro-F1, and the forefoot-only pair (0.435) is worse than any pair retaining a rear or lateral channel, so anatomical coverage dominates sensor count. We retain MET1 as a frequently reported clinical ulceration site [1], [2] at the cost of one ADC channel; Fig. 2 summarises both scales.

D. Robustness, Alert Thresholds and Deployment

Robustness uses a grouped 25% subject holdout estimating R_{hold} with clean baseline 0.847. This is a different estimand from the out-of-fold 0.885 and the two should not be differenced (Table III, Fig. 3). Degradation is graded for faults produced by the radio link and abrupt for faults produced by the transducer: a dead heel channel and a +15 kPa offset each destroy the model, the latter by the occupancy-saturation mechanism derived in (6), which Remark 1 shows is removable. Halving the effective rate to 12.5 Hz was nearly free (0.843) whereas 6.25 Hz was costly (0.714), which implicates peak estimation rather than the rate-invariant PTI: a 0.6 s stance phase receives roughly eight samples at 12.5 Hz but fewer than four at 6.25 Hz, making P_i^* biased low by an amount that grows as the peak narrows.

TABLE III

ROBUSTNESS AND SAMPLING-RATE OPERATING POINTS ON A GROUPED 25% SUBJECT HOLDOUT. PACKET LOSS IS SIMULATED BY ZEROING TIME SAMPLES AND IS AN UPPER BOUND ON DEGRADATION RATHER THAN AN OVER-THE-AIR MEASUREMENT (SEC. III-D).

Perturbation	Severity	Perturbed	Δ	Rel.
Gaussian noise	12 kPa SD	0.641	-0.206	-24%
Dropped packets	30%	0.631	-0.216	-26%
Missing channel	HEEL = 0	0.218	-0.629	-74%
Constant bias	+15 kPa	0.181	-0.666	-79%
Rate reduction	50% (12.5 Hz)	0.843	-0.004	-0.5%

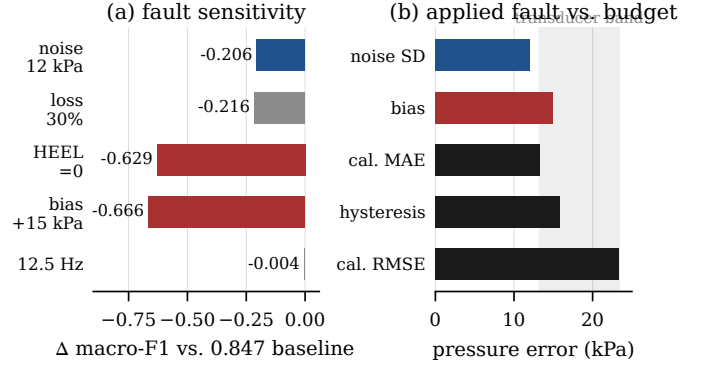


Fig. 3. (a) Fault sensitivity against the 0.847 clean holdout baseline. (b) The magnitude of the applied faults compared with the transducer's own pressure-equivalent error budget from Sec. II-C; the shaded band spans calibration MAE to RMSE. The two faults that were injected lie inside the band, which limits what the robustness curves can be transferred to.

Sweeping the alert cut-off gave sensitivity/specificity 0.844/0.316 at 40 kPa, 0.681/0.933 at 55 kPa and 0.427/0.986 at 75 kPa: the shipped default is deliberately specificity-weighted, a design choice about alert fatigue and not a screening-sensitivity claim. Host-side, feature extraction averaged 0.16 ms and inference 0.07 ms for a combined 0.23 ms path (P95 0.31 ms, $n = 200$), a duty cycle of 0.6% against the 40 ms sample period. BLE airtime, over-the-air error rate and battery life were not measured and are not estimated here.

E. Sparse Versus Dense Sites in Human Walking

Because no X-Step walking data exist, the sparse-sampling assumption was probed on the 32-cell archive by comparing single anatomical cells against the maximum of the corresponding dense cluster (298 foot-takes). Median time-series correlation was 0.986 at MET2, 0.885 at the heel, 0.879 at MET1 and 0.717 at MET5. Pearson correlation is invariant under affine rescaling, so these values certify waveform *shape* and are silent on magnitude; the peak-to-peak correlations of 0.43–0.74 are the relevant statistic, and proper limits of agreement [12] are unavailable across hardware reporting different native units without a cross-calibration we do not have.

A four-site reconstruction of the centre of pressure, $\hat{c}(t) = \sum_i \mathbf{x}_i p_i(t) / \sum_i p_i(t)$, tracked the anteroposterior component (median $r = 0.805$) but not the mediolateral one ($r = 0.005$). This is expected geometrically as well as empirically.

Since $\hat{c}(t)$ is a convex combination of the site coordinates, $\hat{c}(t) \in \text{conv}\{\mathbf{x}_i\}$ for all t , so for any reference point $\bar{\mathbf{x}}$ the reconstructed excursion satisfies $\|\hat{c}(t) - \bar{\mathbf{x}}\| \leq \max_i \|\mathbf{x}_i - \bar{\mathbf{x}}\|$: a four-point measure systematically contracts the true excursion, and with no medial–lateral pair in the midfoot the mediolateral coordinate is nearly fixed by the site geometry. None of these values enter the classification tables.

V. DISCUSSION AND THREATS TO VALIDITY

Three evidence levels should be kept apart. Level A is physical engineering evidence: calibration, timing, packet accounting, latency. Level B is algorithmic evidence on a stated dataset. Level C is clinical evidence: incident ulceration, prevented events, decision impact. This paper contributes Level A and Level B, plus a Level A/B sparse-versus-dense comparison on different hardware. It contributes nothing at Level C: a grouped macro-F1 of 0.885, or 2.31 bits, is a statement about a generator and not about a patient.

The dominant threat is to *construct* validity. Simulator zone labels are a deterministic function of gait class, and peak-only features reached 0.886 against 0.885 for the full 59-dimensional vector, so the generator is largely peak-separable and the problem posed here is narrower than the one a wearer presents; the 2.31-bit figure is a property of that generator’s label entropy, not of human gait. *Internal* validity is protected by grouped splitting and automated leakage tests, but the robustness grid, the ablation lattice and the threshold sweep were all computed after the production head was frozen, which is the correct order yet still selects an operating point on the same generator it is evaluated on. *Statistical-conclusion* validity is limited by 24 clusters: the cluster bootstrap of Sec. III-C has little resolution below the $\delta = 0.02$ margin, which is precisely why equivalence is tested rather than asserted, and why the Fano column of Table II is reported as a bound rather than an estimate. *External* validity is weakest: four resistive sensors cannot reconstruct shear or a full pressure map, drift and creep are not bench-quantified, radio airtime and battery life are unmeasured so no wear-time claim is made, and test–retest reliability characterises the generator only. An unpaired ulcer-image classifier in the repository is excluded, since pairing it with insole data would imply a fusion result we cannot support, and no ethics approval accompanies the reused archive.

Three findings nevertheless generalise as design guidance. Evaluation design dominates apparent performance at this scale. Sparse layouts fail asymmetrically, so sequence-gap accounting and per-channel plausibility checks matter more than a smoothly degrading score, which will silently mislead when an electrode lifts. And redundancy, not rank, is the right frame for site selection: the site whose removal is cheapest may be the one whose partner is doing its work, a conclusion that only becomes visible once layouts are scored as coalitions rather than ranked individually.

VI. CONCLUSION

X-Step specifies a four-site plantar-pressure insole with a documented 28-byte BLE contract, operator-attested transducer calibration placed in an explicit error budget, a leakage-controlled modelling stack with pre-specified estimands and subject-cluster inference, and ablation, robustness and host-latency operating curves. Under subject-grouped evaluation the production head reaches macro-F1 0.885—at least 2.31 of 3.17 bits of label information—at 7.7 kB and 0.23 ms of host path. It degrades gracefully under noise, bounded packet loss and halved sampling rate, and fails outright under channel loss or constant bias, the latter predictable from the threshold features themselves and removable by per-window re-baselining. Scored as a cooperative game, the four sites are nearly additive in bits but strongly interacting in pairs, so MET5 and HEEL carry separate information while the forefoot pair is internally redundant and jointly load-bearing. Offset-equivariant features, measured radio and power figures, bench characterisation of drift over wear cycles, and a prospective walking study under ethics approval are the bridge to any patient-facing statement.

REFERENCES

- [1] D. G. Armstrong, A. J. M. Boulton, and S. A. Bus, “Diabetic foot ulcers and their recurrence,” *N. Engl. J. Med.*, vol. 376, no. 24, pp. 2367–2375, 2017.
- [2] M. E. Fernando *et al.*, “Plantar pressure in diabetic peripheral neuropathy patients with active foot ulceration, previous ulceration and no history of ulceration: a meta-analysis of observational studies,” *PLoS ONE*, vol. 9, no. 6, p. e99050, 2014.
- [3] N. C. Schaper *et al.*, “Practical guidelines on the prevention and management of diabetes-related foot disease (IWGDF 2023 update),” *Diabetes Metab. Res. Rev.*, vol. 40, p. e3657, 2024.
- [4] S. A. Bus *et al.*, “Footwear and offloading interventions to prevent and heal foot ulcers and reduce plantar pressure in patients with diabetes: a systematic review,” *Diabetes Metab. Res. Rev.*, vol. 32, no. S1, pp. 99–118, 2016.
- [5] P. R. Cavanagh and S. A. Bus, “Off-loading the diabetic foot for ulcer prevention and healing,” *J. Vasc. Surg.*, vol. 52, no. 3S, pp. 37S–43S, 2010.
- [6] A. H. Abdul Razak, A. Zayegh, R. K. Begg, and Y. Wahab, “Foot plantar pressure measurement system: a review,” *Sensors*, vol. 12, no. 7, pp. 9884–9912, 2012.
- [7] N. Hegde, M. Bries, and E. Sazonov, “A comparative review of footwear-based wearable systems,” *Electronics*, vol. 5, no. 3, p. 48, 2016.
- [8] N. Hegde and E. Sazonov, “SmartStep: a fully integrated, low-power insole monitor,” *Electronics*, vol. 3, no. 2, pp. 381–397, 2014.
- [9] D. G. Armstrong, K. Holtz-Neiderer, C. Wendel *et al.*, “Skin temperature monitoring reduces the risk for diabetic foot ulceration in high-risk patients,” *Am. J. Med.*, vol. 120, no. 12, pp. 1042–1046, 2007.
- [10] S. Holm, “A simple sequentially rejective multiple test procedure,” *Scand. J. Stat.*, vol. 6, no. 2, pp. 65–70, 1979.
- [11] C. Nadeau and Y. Bengio, “Inference for the generalization error,” *Mach. Learn.*, vol. 52, no. 3, pp. 239–281, 2003.
- [12] J. M. Bland and D. G. Altman, “Statistical methods for assessing agreement between two methods of clinical measurement,” *Lancet*, vol. 327, no. 8476, pp. 307–310, 1986.
- [13] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Hoboken, NJ, USA: Wiley, 2006.
- [14] “Multimodal gait dataset: synchronized sensorized-insole (pressure and inertial) and optical motion-capture recordings,” Zenodo, 2026. doi: 10.5281/zenodo.20156243.